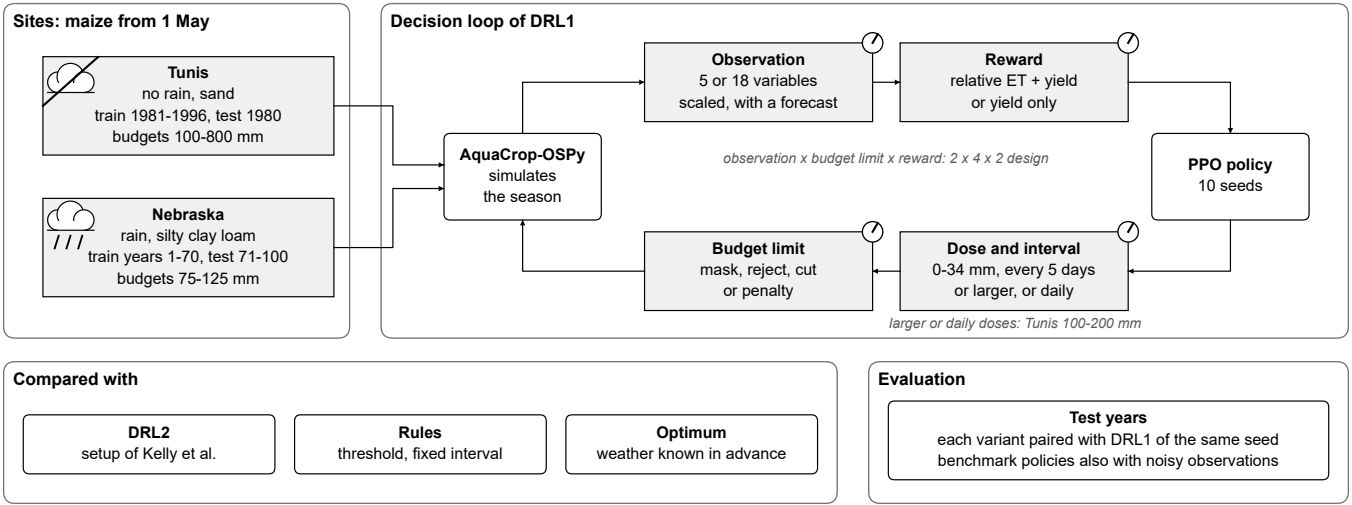

Sites: maize from 1 May
Tunis
no rain, sand
train 1981-1996, test 1980
budgets 100-800 mm
Nebraska
rain, silty clay loam
train years 1-70, test 71-100
budgets 75-125 mm
Decision loop of DRL1
AquaCrop-OSPy
simulates
the season
Observation
5 or 18 variables
scaled, with a forecast
Reward
relative ET + yield
or yield only
observation x budget limit x reward: 2 x 4 x 2 design
PPO policy
10 seeds
Budget limit
mask, reject, cut
or penalty
Dose and interval
0-34 mm, every 5 days
or larger, or daily
larger or daily doses: Tunis 100-200 mm
Compared with
DRL2
setup of Kelly et al.
Rules
threshold, fixed interval
Optimum
weather known in advance
Evaluation
Test years
each variant paired with DRL1 of the same seed
benchmark policies also with noisy observations